INCH-POUND

MIL-H-87041B(NU)
26 February 1990
SUPSERSEDING
MIL-H-87041A(NU)
28 October 1983

MILITARY SPECIFICATION

HAT, SERVICE (WHITE)

This specification is approved for use by the Navy Clothing and Textile Research Facility, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for white service hats for use by Navy enlisted men and midshipmen.

* 1.2 Classification. The service hats shall be of the following types and sizes (see 6.2):

Type I - Hat, Enlisted Man's
Type II - Hat, Midshipman's

Schedule of Sizes

6 1/2, 6 3/4, 7, 7 1/4, 7 1/2, 7 3/4, 8, 8 1/4, and 8 1/2

2. APPLICABLE DOCUMENTS

* 2.1 Government documents.

* 2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Officer in Charge, Navy Clothing and Textile Research Facility, 21 Strathmore Road, Natick, MA 01760-2490, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8405

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:52Z
Check the source to verify that this is the current version before use.

SPECIFICATIONS

FEDERAL

V-T-276	- Thread, Cotton
NN-P-71	- Pallet, Material Handling, Wood, Stringer Construction, 2 Way and 4 Way (Partial)
CCC-C-419	- Cloth, Duck, Cotton, Unbleached, Plied-Yarns, Army and Numbered
CCC-C-430	- Cloth, Sheeting, Cotton, and Polyester and Cotton
CCC-C-438	- Cloth, Buckram, Woven and Nonwoven
CCC-C-461	- Cloth, Twill, Uniform, Cotton
DDD-L-20	- Label; for Clothing, Equipage, and Tentage (General Use)
DDD-T-86	- Tape, Textile, Cotton, General Purpose (Unbleached, Bleached or Dyed)

MILITARY

MIL-P-15011	- Pallet, Material Handling, Wood, Post Construction, 4 Way Entry
MIL-B-17757	- Boxes, Shipping, Fiberboard (Modular Sizes)
MIL-T-43548	- Thread, Polyester Core; Cotton-, Rayon- or Polyester-Covered

STANDARDS

FEDERAL

FED-STD-191	- Textile Test Methods
FED-STD-751	- Stitches, Seams, and Stitchings

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	- Marking for Shipment and Storage
MIL-STD-147	- Palletized Unit Loads
MIL-STD-2073-1	- DoD Materials Procedures Development and Application of Packaging Requirements
MIL-STD-2073-2	- Packaging Requirement Codes

(Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

* 2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

LAWS AND REGULATIONS

U.S. Postal Manual

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:52Z
Check the source to verify that this is the current version before use.

(Copies of the manual should be addressed to the Superintendent of Documents, U.S. Government Printing Office. Washington, DC 20402-0001.)

* 2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Applications for copies should be addressed to the American Trucking Association, ATTN: Traffic Department, 1616 P Street, N.W., Washington, DC 20036-1404.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Applications for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, Illinois 60606-5808.)

THE COLOR ASSOCIATION OF THE UNITED STATES

Department of Defense Standard Shades For Sewing Threads

(Applications for copies should be addressed to the Color Association of The United States, 343 Lexington Ave. New York, NY 10016-0927.)

TECHNICAL MANUAL OF THE AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS

Method No. 8 - Colorfastness to Crocking: AATCC Crockmeter Method

(Applications for copies should be addressed to AATCC National Headquarters, P.O. Box 12215, Research Triangle Park, NC 27709-2215.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

* 2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Guide sample. Samples, when furnished, are solely for guidance and information to the contractor (see 6.3). Variations from this specification may appear in the sample, in which case this specification shall govern.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:52Z
Check the source to verify that this is the current version before use.

* 3.2 First article. When specified, (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.3.

3.3 Materials.

* 3.3.1 Cloth, twill, cotton. The basic material used for the types I and II hats shall be white, shade 3035, Standard Sample Piece Number 84148 and shall conform to types I, II, III or IV of CCC-C-461 with the following exceptions:

a. The weight per square yard shall be 7.2 oz. minimum (no maximum).

b. The dimensional stability shall not exceed an average of 2% in either the warp or filling directions with no individual sample unit greater than 2.5%.

c. The minimum breaking strength in the filling shall be 95 lbs.

3.3.2 Cloth, duck, cotton. The cloth used for the interlining of the hat brim shall conform to type I, number 6 or 8, hard texture, of CCC-C-419, except that air permeability requirements shall not apply.

* 3.3.3 Cloth, buckram, cotton. The cloth used for the reinforcement of the brim interlining shall be bleached white and shall conform to type III, class 1 of CCC-C-438.

* 3.3.3.1 Cloth, nonwoven, polyester and rayon. As an alternate to the materials specified in 3.3.2 and 3.3.3, a 70% polyester/30% rayon nonwoven material, cut as specified in 3.5 and conforming to the following requirements may be used (see 4.4.1). The material shall conform to Pellon style number 3790 or equal (see 6.4 and 6.5).

Binder - Acrylic		
Weight per square yard, oz.	-	4.7 (± 0.4)
Breaking strength, lbs.		
	- Warp	45 (min)
	- Filling	45 (min)

3.3.4 Cloth, sheeting, cotton and polyester and cotton. The cloth used to bind the bottom edge of the crown shall be bleached white and shall conform to type II, class 2, style A, B or F of CCC-C-430 or the twill cloth requirements of 3.3.1. As an alternate, a bleached cotton or polyester/cotton cloth woven with a plain or twill weave conforming to the following requirements, when tested as specified in 4.4.1, may be used. The cloth shall be bias-cut into strips 1 1/8 to 1 3/8 inches wide.

Weight per square yard, oz.	-	4.0 (min)
Yarns per inch	- Warp	60 (min)
	- Filling	50
Breaking strength, lbs.		
	- Warp	50 (min)
	- Filling	45
Shrinkage, percent		
	- Warp	2.0 (max)
	- Filling	2.0

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:52Z
Check the source to verify that this is the current version before use.

* 3.3.5 Tape, woven. The tape used for the hanger loops shall be cotton, 1/2 or 5/8 inch wide, bleached white and shall conform to type II, class 2 of DDD-T-86, except that the filling yarns per inch shall be 55 (minimum). As an alternate, a 1/2 to 5/8 inch bleached white tape having a minimum warp strength of 50 lbs. may be used (see 4.4.1).

* 3.3.6. Thread. The thread used for general seaming and stitching shall be cotton, type 1A2, 3 ply, ticket no. 70 conforming to V-T-276. The thread shall be shade White AH, Cable No 66050, in accordance with the Department of Defense Standard Shades For Sewing Threads. The cotton thread used for stitching the blue binding to the brim of the type II hat shall be type 1A3, 3 ply, ticket no. 70 conforming to V-T-276. The thread shall be dyed shade Black, AA, Cable No. 66043, in accordance with the Department of Defense Standard Shades For Sewing Threads. As an alternate to the cotton thread, a covered polyester thread conforming to ticket no. 70, 2 or 3 ply, of MIL-T-43548. may be used. The polyester core polyester-covered thread shall not be used.

* 3.3.6.1 Colorfastness. (Type II hat) The dyed thread shall show colorfastness to light, perspiration, and combined laundering and bleaching equal to or better than the standard sample when tested as specified in 4.4.1. As a limit of acceptability or when no standard sample is available, the dyed thread shall show a minimum of "good" fastness to light, perspiration, and combined laundering and bleaching when tested as specified in 4.4.1.

3.3.7 Cloth, binding. (Type II hat) The cloth used for binding the brim of the Type II hat shall be the same as the material specified in 3.3.1, except that it shall be dyed to match shade Blue 3362.

* 3.3.7.1 Colorfastness. The dyed cloth shall show colorfastness to light, perspiration and combined laundering and bleaching equal to or better than the standard sample and an AATCC reading for crocking not lower than 3.0 when tested as specified in 4.4.1. As a limit of acceptability, or when no standard sample is available, the dyed cloth shall show a minimum of "good" colorfastness to light, perspiration, combined laundering and bleaching, and an AATCC reading for crocking not lower than 3.0 when tested as specified in 4.4.1.

* 3.3.8 Marking. The contract number, applicable hat size and laundering instructions shall be legibly marked on the sweatband approximately 3/8 inch to the left or right of the back joining seam. As an option, the hat size and contract number may be marked on one side of the hanger loop, with the laundering instructions on the opposite side. The markings shall conform to type III or IV, class 4 of DDD-L-20 for laundered items, except that the inscription shall read as follows:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:52Z
Check the source to verify that this is the current version before use.

(Sweatband only)

DLA100-00-0-0000 SIZE 7 1/4 (Example)
HAND OR MACHINE WASH
WARM WITH MILD DETERGENT.
TUMBLE DRY LOW OR HANG DRY

(Hanger loop only)

DLA100-00-0-0000 HAND OR MACHINE WASH WARM WITH MILD DETERGENT	S TUMBLE DRY LOW I OR HANG DRY Z E

The finished length of the label shall be 2 to 2 1/8 inches (see Figure 2).

3.4 Design. Type I and Type II hats shall be made of a white twill fabric and shall have a 6 piece dome shaped crown with a flared stitched turnup brim and self fabric sweatband. The top edge of the brim of the Type II Midshipman's hat shall be bound with a blue dyed self fabric (see Figure 1).

3.4.1 Figure. The figure is furnished for information purposes only. When inconsistencies exist between the specification and the figure, the specification shall govern.

* 3.5 Patterns and blocks. Standard patterns will be furnished by the Government to the contractor for use in cutting working patterns (see 6.3). The working patterns shall be identical to the Government patterns. Neither the Government patterns not the working patterns shall be altered in any way, except that additional notches for use during construction are allowed on the working patterns. Also, minor modifications are permitted where necessary when using automatic equipment. These modifications shall not alter the dimensional, serviceability, or appearance requirements cited in the specification. The seam allowance shall be as specified. Each hat shall be properly blocked on commercial forms which shall be furnished by the contractor.

Material	Nomenclature of pattern parts	Cut parts
Cloth, twill, cotton	Brim	1
	Crown, center side	2
	Crown, left of center	2
	Crown, right of center	2
	Sweatband	1
Cloth, duck, cotton	Brim (interlining 1/)	2
Cloth, buckram, cotton	Brim (interlining 1/ 2/)	1

1/ When the alternate polyester/rayon material is used (see 3.3.3.1) four (4) parts shall be cut using the cotton duck pattern piece. The pattern piece for the cotton buckram is not used with the alternate interlining.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:52Z
Check the source to verify that this is the current version before use.

2/ This one size pattern is supplied as a guide for grading the complete size range. (see operation 1.c (2)).

* 3.6 Construction.

* 3.6.1 Stitches, seams, and stitchings. Stitch, seam, and stitching types specified in Table I shall conform to FED-STD-751. Whenever two or more methods for seam or stitch types are given for the same operation, any one of them may be used. Seam allowances shall be maintained with seams sewn so that no raw edges, runoffs, twists, pleats, puckers or open seams result. All seams shall start and finish evenly. Thread tension shall be maintained so that there is no tight or loose stitching. Unless otherwise specified, topstitching and edgestitching shall be 1/16 to 1/8 inch from the edge.

3.6.2 Stitches per inch. The minimum and maximum number of stitches per inch shall be specified in Table I.

* 3.6.3 Thread breaks and ends of seams. Ends of all seams and stitchings when not caught in other seams or stitchings shall be backtacked not less than 3/8 inch. The ends of a continuous line of stitching shall be overlapped not less than 1/2 inch. Thread breaks (all stitch types) shall be secured by stitching back of break not less than 1/2 inch.

3.7 Manufacturing operations requirements. The hats shall be manufactured in accordance with the operations specified in Table I. The contractor is not required to follow the exact sequence of operations listed provided that the finished hats are identical to those produced by following the sequence specified in Table I. All pressing and blocking operations referenced shall be accomplished with a heated pressing iron or machine after all sewing and mechanical operations on the hat have been completed. Minor modifications are permitted where necessary when using automatic equipment. These modifications shall not alter the dimensional, serviceability, or appearance requirements cited in the specification.

3.7.1 Shade and size marking. The component parts shall be marked, ticketed, or bundled to insure a uniform shade and size throughout the hat. Any method may be used except:

1. Corrosive metal fastening devices.
2. Sew-on tickets.
3. Adhesive type tickets whose adhesive causes discoloration or adheres to the material upon removal of tickets.

The use of an ink pad numbering machine, rubber stamp or pencil is acceptable provided the numbers are legible and are not visible on the outside of the hat.

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:52Z
Check the source to verify that this is the currentversion before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:52Z
Check the source to verify that this is the current version before use.

TABLE I - CONSTRUCTION OF HATS

NO.	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH IN	THREAD NDL	THREAD BOB/ LPR
1.	Cutting. a. The hats shall be cut in strict accordance with patterns which shall show size, shape, and directional lines for cutting, and notches for proper assembly of all component parts. b. Crown component parts shall be cut from one piece of material. The brim twill cloth (see 3.3.1) shall be a good shade match to the crown. c. Optional cutting of brim twill, brim buckram and cover strips - In addition to specified cutting operations 1a and 1b, the following alternate cutting methods and preparations may be used: (1) Brim twill - To insure maximum use of various widths of material, the pointed angle at each end of the pattern may be varied between 30 and 54 degrees. The angle cut at each end of the pattern shall be the same.					
*	(2) Brim buckram - Cut the buckram interlining (when applicable) on a 45 (±5) degree bias in strips a minimum of 3 inches wide. The ends of the strips shall be butted (not overlapped) and joined with a stitch bight (throw) of 1/8 to 1/4 inch.	Any 500 or 600 Series	FSa-1	8-10	Comm.	
	(3) Crown seam covering - The twill cloth (see 3.3.1) shall be cut on the bias into strips 1 1/8 inches wide. The strip ends shall be joined together 1/8 to 3/16 inch from raw edges and the seams shall be pressed open.	301	SSa-1	10-12	70	70
*	(4) Brim binding (Type II only). See operation 6 for cutting instructions.					
*	(5) Sweatband. See operation 8a for cutting instructions.					

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:52Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:52Z
Check the source to verify that this is the current version before use.

TABLE I - CONSTRUCTION OF HATS

NO.	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH IN	THREAD NDL	THREAD BOB/ LPR
5.	Make brim.					
	a. Stitch ends of twill brim covering 5/16 to 7/16 inch from raw edge forming a continuous circle.	301	SSa-1	10-12	70	70
	b. Press seam open and flat.					
*	c. Fold circular twill brim covering in accordance with fold line on pattern, back side out. Exercising great care, align the folded edge of twill brim covering with top edge of brim interlining on the inner side of interlining and join together, 1/8 to 3/16 inch from top edges of brim.	101 or 301	SSa-1	6-8	70 70	 70
	d. Turn brim covering over top edge of interlining to finished position. The seamline shall finish on the inner side of the brim facing the crown in the finished hat.					
	e. Sew one or more rows of stitching through all plies of brim to hold the brim covering and interlining in place for subsequent multiple stitching operations. Any of the following methods may be used:					
	(1) One near top edge below joining seamline.	301	SSv-1	10-12	70	70
	(2) One near top edge below joining seamline, one near center, and one near bottom edge.	301	SSv-1	10-12	70	70
	(3) One near top edge below joining seamline and one or more near bottom edge.	301	SSv-1	10-12	70	70
	(4) One near bottom edge.	301	SSv-1	10-12	70	70
	(5) Any combination of the above.	301	SSv-1	10-12	70	70
	f. Stitch 70 to 75 continuous rows of evenly spaced and parallel stitching to the entire circumference of the brim, starting 1/4 to 3/8 inch from top folded edge and terminating beyond the bottom edge of interlining canvas.	301	SSv-1	10-12 (70-75 rows)	70	70

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:52Z
Check the source to verify that this is the current version before use.

TABLE I - CONSTRUCTION OF HATS

NO.	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH IN	THREAD NDL	THREAD BOB/ LPR
	NOTE: When sweatband is already attached the rows of stitching shall extend beyond the edge of sweatband.					
	g. The finished stitched brim shall measure 2 1/2 to 2 7/8 inches in width from top folded edge of brim to breakline at bottom of brim. The center of the diagonal joining seam shall finish ± 1 inch at center back of the finished hat.					
6.	Bind top edge of brim (Type II hat only).					
*	a. The binding for the brim shall be made of twill cloth as specified in 3.3.7. Cut a 2 3/4 inch wide bias cut strip of cloth of sufficient length for each size.					
	b. Binding shall extend 7/8 to 1 1/8 inches on each side of brim with joining seam of binding finishing at back of hat.					
*	c. Position the binding on the inner side of the brim, 3/4 to 7/8 inch down from the top folded edge of brim. Starting at center back of brim, with end of binding folded over 1/2 inch, stitch binding to brim, 1/8 to 1/4 inch from raw edge of binding, completely around brim with the binding extended over folded end 1/2 to 3/4 inches.	101 or 301	BSo-2(a)	10-12	70 70	 70
	NOTE: The binding shall not be spliced.					
	d. Turn binding up over the top of brim, fold under raw edge and stitch opposite edge of binding to outer side of brim 1/16 to 1/8 inch from folded edge.	301	BSo-2(b)	10-12	70	70
	e. Stitch 5 to 7 rows of stitching on binding through all plies of material.	301	SSv-1 5-7 rows	10-12	70	70

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:52Z
Check the source to verify that this is the current version before use.

TABLE I - CONSTRUCTION OF HATS

NO.	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH IN	THREAD NDL	THREAD BOB/ LPR
7.	Make crown.					
	a. The crown shall consist of six pieces.					
	NOTE: Notches in patterns shall be used as a guide only to indicate proper assembly points of crown parts. Cut notches are permitted in the seam allowance of the crown pieces, but the notch depth shall not exceed 3/32 inch.					
	b. Seam the adjacent sides of the crown pieces together, face to face, 3/16 to 5/16 inch from the raw edge, matching notches accordingly to form the crown.	301 *or* 401	SSag-3(a)	10-12	70	70
	c. Press seams open and flat.					
	or					
	As an alternate to pressing, the seam allowances may be spread open mechanically and held flat by the cover stripping applied in operation 7d.					
*	d. Double stitch cover strips made of the basic twill cloth on all crown seams, 3/8 to 7/16 inch gage with rows of stitching 1/16 inch from folded edges. Turn under evenly the edges of stripping to insure that the finished width of each cover strip measures 5/8 ± 1/16 inches and that the seamline to be covered is positioned equidistant by ± 1/8 inch from finished folded edges of stripping. See operation 1.C(3).	301	SSag-3(b)	10-12	70	70
*	NOTE: There shall be no more than one splice per cover strip.					

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:52Z
Check the source to verify that this is the current version before use.

TABLE I - CONSTRUCTION OF HATS

NO.	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH IN	THREAD NDL	THREAD BOB/ LPR
*	e. When attaching last cover stripping to crown, simultaneously insert and catch a hanger loop in the line of stitching. The loop shall be located midway between top and bottom of crown and towards the back of the hat. The single loop shall be formed by doubling a piece of tape (see 3.3.5) cut 4 to 4 1/4 inches long. Finished loop shall measure 2 to 2 1/8 inches long.	301	SSa-1	10-12	70	70
	f. Stitch a bias-cut strip of sheeting specified in 3.3.4 around outer bottom edge of crown; turn and stitch binding to inside edge of crown, with ends of binding overlapping themselves by not less than 1/2 inch.	101 or 301	BSf-2	8-10	70 70	70
	g. In lieu of operation 7f, crown binding may be performed by stitching a bias-cut strip of sheeting to the edge of the crown, with the turned edge of the binding on the outer side of the crown and with the ends of binding overlapping themselves by not less than 1/2 inch.	101 or 301	BSb-1	8-10	70 70	70
	NOTE: When performing this alternate operation care shall be exercised to avoid gathering of material under tension.					
	h. The finished binding shall measure 1/4 to 1/2 inch wide.					
8.	Make sweatband. a. The sweatband shall be made of the basic twill cloth specified in 3.3.1 and shall be straight cut in one piece 1 5/8 to 1 3/4 inches in width.					
	b. Turn under both edges 1/4 to 3/8 inch and stitch 1/16 to 1/8 inch from each folded edge.	101 or 301 or 401	EFa-1	8-10	70 70	70
	or					

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:52Z
Check the source to verify that this is the current version before use.

TABLE I - CONSTRUCTION OF HATS

NO.	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH IN	THREAD NDL	THREAD BOB/ LPR
	c. In lieu of operation 8b, sweatband may be formed by turning under both edges 1/4 to 3/8 inch and heat pressed or mechanically creased, without stitching.					
	d. The finished sweatband shall measure 7/8 to 1 1/8 inches in width.					
	e. Stitch ends of sweatband together forming a continuous circle, 1/4 inch from raw edge.	301	SSa-1	10-12	70	70
	NOTE: Finished length of sweatband shall equal head circumference for hat sizes, as indicated in Table II herein.					
9.	Hat assembly.					
	a. Stitch bottom bound edge of crown to inner edge of brim. The distance from the outer edge of brim to bound edge of crown shall be 2 13/16 to 2 15/16 inches.	301	LSa-1	10-12	70	70
	b. Stitch sweatband to inside of hat on brim with folded edge of sweatband 3/16 to 3/8 inch from edge of the covered interlining. The joining seam of the sweatband shall finish approximately 1 inch from center back crown seam. The distance from the outer edge of brim to bottom folded edge of sweatband shall be 2 13/16 to 2 15/16 inches.	301	LSd-1	10-12	70	70
*	c. Stitch free edge of sweatband, 1/16 to 1/8 inch from folded edge, to inside of crown. Continue stitching sweatband to crown with 8 to 10 rows of stitching, uniformly spaced along the width of the sweatband.	301	LSd-1 8 to 10 rows	8-12	70	70
10.	Marking.					
*	The contract number, size and laundering instructions shall be centered and marked on sweatband toward back of hat or marked on hanger loop (see 3.3.8).					

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:52Z
Check the source to verify that this is the current version before use.

TABLE I - CONSTRUCTION OF HATS

NO.	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH IN	THREAD NDL BOB/ LPR
11.	Blocking. Block hat neatly on the correct size block utilizing live steam and heated block. As an alternate, dry heat hydraulic blocking is permitted.				
12.	Cleaning. a. Trim and remove all loose ends of thread on the inside and outside of the hat. b. Remove all spots and stains. c. Remove all shade tickets. NOTE: Care shall be taken when trimming thread ends so that damage to the fabric does not result. The use of chalk, powder, etc., to cover spots and stains is prohibited.				

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:52Z
Check the source to verify that this is the current version before use.

* 3.8 Sizes and measurements. The sizes and measurements of the finished hats shall be as shown in Table II. All measurements and tolerances are expressed in inches.

Table II - Finished Sizes and Measurements

Hat (marked) size	Head circumference inches	Hat (marked) size	Head circumference inches
6 1/2	20 1/2	7 1/2	23 1/2
6 3/4	21 1/4	7 3/4	24 1/4
7	22	8	25
7 1/4	22 3/4	8 1/4	25 3/4
		8 1/2	26 1/2
Tolerance	+ 3/8 - 1/4 inch	Tolerance	+ 3/8 - 1/4 inch

NOTE: A standardized head size gauge will be loaned by the Government to the contractor for determination of head circumference.

3.9 Workmanship. The finished hats shall conform to the quality established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

* 4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to acertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Classification of inspection. The inspection requirements specified

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:52Z
Check the source to verify that this is the current version before use.

herein are classified as follows:

1. First article inspection (see 4.3).
2. Quality conformance inspection (see 4.4).

4.3 First article inspection. The first article, submitted in accordance with 3.2, shall be inspected for compliance with design, construction, workmanship, appearance and dimensional requirements.

4.4 Quality conformance inspection. Inspection shall be in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated herein.

* 4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document. In addition, a certificate of compliance shall be submitted for the nonwoven fabric requirements of 3.3.3.1, the binding requirements of 3.3.4 and for the alternate woven tape requirements of 3.3.5. The methods of testing specified in FED-STD-191 and as listed in Table IV shall be followed. A component lot shall be unacceptable if one or more sample units fail to meet any of the test requirements specified. The sample sizes for testing shall be as follows:

Table III - Sample Sizes

Lot size (yards)	Sample unit
800 or less	2
801 thru 22,000	3
22,001 and over	5

Table IV - Instructions for testing

Component	Characteristic	Requirement	Test Method
Thread, black	Colorfastness to:		
	Light	3.3.6.1	5660
	Perspiration	3.3.6.1	5680
	Combined laundering and bleaching	3.3.6.1	5605
Cloth, blue	Colorfastness to:		
	Light	3.3.7.1	5660
	Crocking	3.3.7.1	AATCC - 8
	Perspiration	3.3.7.1	5680
	Combined laundering and bleaching	3.3.7.1	5605

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:52Z
Check the source to verify that this is the current version before use.

4.4.2 Examination of the end item. The defects found during the examination of the end item shall be classified in accordance with 4.4.2.1 and 4.4.2.2. The applicable inspection levels and acceptable quality levels (AQL's) shall be as indicated in 4.4.2.3. The sample unit shall be one completely fabricated hat. The lot size shall be expressed in units of hats.

4.4.2.1 Visual examination. The finished hats shall be examined for defects in design, material, construction, workmanship, and marking and the defects classified in accordance with the list below:

Defects	Classification Major	Minor A	Minor B
I MATERIAL DEFECT(S) AND DAMAGES(S)			
Any size hole, cut, tear, smash, misweave, needle chews, visible mend or burn	X		
II SHADED PART(S)			
a. Marked variation in shade between two or more crown parts on outside	X		
b. Marked variation in shade within a part on outside	X		
c. Marked variation in shade between brim and crown	X		
d. Marked variation in shade between seam covering tape and crown parts on inside, affecting appearance on outside		X	
III CLEANNESS			
a. Any spot or stain clearly noticeable:			
(1) - on outside of brim or crown (brim up)	X		
(2) - on inside of brim (brim up)		X	
(3) - on inside of crown:			
- affecting outside	X		
- not affecting outside			X
b. More than three thread ends not trimmed to 1/4 inch or less on outside (including both sides of brim)		X	
c. More than five thread ends not trimmed to 1/2 inch or less on inside			X
d. Adhesive ticket or any exposed marking not removed			X
IV COMPONENT(S) AND ASSEMBLY			
a. Any component part or operation omitted (unless otherwise classified herein)	X		

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:52Z
Check the source to verify that this is the current version before use.

Defects	Classification Major	Minor A	Minor B
b. Any component not as specified (type, size, color, etc.)	X		
c. Any component part tight, twisted, distorted, pleated or full (unless otherwise classified herein)		X	
d. Any operation not as specified (unless otherwise classified herein)		X	
V CUTTING			
Any part not cut in accordance with directional lines indicated on patterns or in accordance with specification requirements	X		
VI PRESSING AND BLOCKING			
a. Pressing and blocking omitted	X		
b. Seams not pressed as required			X
c. Poorly blocked		X	
VII SEAMS AND STITCHING			
a. General			
1. Any part of hat caught in any unrelated stitching operation	X		
2. Seam twisted, puckered or pleated		X	
3. End of seam or stitching, when not caught in another seam or stitching, backtacked less than 3/8 inch			X
4. Stitching overlapping ends of thread break or skipped stitches less than 1/2 inch			X
5. Wrong shade or various shades of thread used on outside of same hat		X	
b. Stitching margin			
1. Irregular, affecting appearance		X	
2. Edge stitching sewn too close to edge, resulting in damage to cloth	X		
3. Beyond range of width specified or varies more than 1/16 inch when no range is specified			X
c. Open seams (skipped or broken stitches)			
1. All seams except inside seam coverings			

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:52Z
Check the source to verify that this is the current version before use.

Defects	Classification Major	Minor A	Minor B
and sweatband:			
a. up to 1/4 inch			X
b. 1/4 inch up to 3/4 inch inclusive		X	
c. more than 3/4 inch	X		
2. On inside seam coverings and sweatband:			
a. up to 3/4 inch inclusive			X
b. more than 3/4 inch		X	
d. Raw edges (on outside of hat)			
1. up to 1/4 inch			X
2. 1/4 inch up to 3/4 inch inclusive		X	
3. more than 3/4 inch	X		
4. any raw edge more than one inch on inside of hat			X
NOTE: One or more broken stitches or two or more continuous skipped or runoff stitches constitute an open seam.			
e. Seam and stitch types			
1. Not specified seam or stitch type		X	
2. Any line of stitching not beginning or ending where specified			X
3. Any required line of stitching omitted	X		
f. Stitch tension			
1. Loose tension, resulting in a loose seam	X		
2. Loose tension, resulting in loosely exposed loops of lower or top thread (score only when condition exists on more than 1/2 the length of seam)		X	
3. Tight tension (stitches break when normal strain is applied in the direction of the seam or stitching)	X		
g. Stitches per inch			
(score only when the condition exists on more than 1/2 the length of seam)			
1. One stitch less than the minimum or any number in excess of the maximum specified			X

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:52Z
Check the source to verify that this is the current version before use.

Defects	Classification Major	Minor A	Minor B
2. Two or more stitches less than the minimum specified		X	
VIII BRIM BINDING (Type II hat only)			
a. Other than specified color	X		
b. Puckered, twisted or distorted	X		
c. Not uniform, extending less than 7/8 inch or more than 1 1/8 inches from top folded edge of brim	X		
d. Joining seam not positioned at back of hat			X
e. Spliced		X	
IX BRIM			
a. Permanently creased or puckered at sweatband joining	X		
b. Puckered, twisted or distorted	X		
c. Less than 3/16 inch or more than 3/8 inch space between end of interlining and bottom edge of sweatband		X	
d. Less than 2 1/2 inches or more than 2 7/8 inches from top edge to break of brim	X		
e. Width of brim irregular, affecting appearance	X		
f. Top edge of brim poorly formed or puckered		X	
g. Rows of continuous stitching not evenly spaced or irregular, affecting appearance	X		
h. Joining seam not positioned at back of hat		X	
X CROWN			
a. Any crown piece sewn in reverse	X		
b. Any seam cover omitted	X		
c. Crown binding less than 1/4 inch or more than 1/2 inch in width			X
d. Any seam cover more than 5/8 inch or less than 1/2 inch in width		X	
e. Hanger loop omitted		X	
f. Hanger loop less than 3/4 inch or more than 1 1/8 inches in length			X
g. Outer edge of crown binding less than 2 13/16 inches or more than 2 15/16 inches from outer edge of brim			X
h. Raw edge of crown seam allowance exposed along outer edge of seam covering tape for more than 1 inch			X
i. Notch cut beyond width of seam allowance,			

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:52Z
Check the source to verify that this is the current version before use.

Defects	Classification Major	Minor A	Minor B
end of notch visible on outside of crown	X		
j. Folded edges of cover strip not positioned equidistant (± 1/8 inch) from seamline			X
XI SWEATBAND			
a. More than one-piece or not straight cut self fabric	X		
b. Puckered, twisted or distorted, affecting proper fit	X		
c. Sweatband irregular in width		X	
d. Width less than 7/8 inch or more than 1 1/8 inches		X	
e. Joining seam of sweatband superimposed on any crown seam covering		X	
f. Distance from outer edge of brim to folded edge of sweatband less than 2 13/16 inches or more than 2 15/16 inches			X
g. Sweatband exposed below folded bottom edge of brim			X
XII MARKING (contract number and size)			
a. Size omitted, incorrect or illegible	X		
b. Contract marking omitted, incorrect or illegible		X	
c. Size and/or contract marking improperly placed			X

4.4.2.2 Dimensional examination. The finished hats shall be examined for defects in measurement in accordance with Table II. Any measurement deviating from the stated requirements shall be scored as a defect (see 4.4.2.3).

4.4.2.3 Inspection levels and acceptable quality levels (AQL's) expressed as defects per 100 units (DHU) for visual and dimensional examinations shall be as follows:

	Inspection level	AQL's
For visual examination in 4.4.2.1		
Major	II	4.0
Major and Minor A (combined)	II	10.0
Total (Major, Minor A and B combined)	II	15.0
For dimensional examination in 4.4.2.2		
One class	S-3	4.0

* 4.4.3 Examination of packaging requirements. An examination shall be made to determine that packaging, packing, and marking comply with Section 5 requirements of this specification. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:52Z
Check the source to verify that this is the current version before use.

prepared for delivery. Defects of closure listed below shall be examined on shipping containers fully prepared for delivery. The lot size shall be on the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL shall be 2.5 defects per hundred units.

Examine	Defect
Marking (exterior and interior	Omitted, incorrect, illegible, of improper size, location, sequence, or method of application.
Materials	Any component missing, damaged, or not as specified.
Workmanship	Inadequate application of components, such as: incomplete closure of container flaps, loose strapping, improper taping, inadequate stapling, bulged or distorted container.
Content	Number of items per shipping container is more or less than required. Size shown on one or more items not as specified on shipping container. 1/

1/ For this defect, one item from each shipping container in sample shall be examined.

* 4.4.4 Palletization examination. An examination shall be made to determine that the palletization complies with the section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one palletized unit load fully packaged. The lot size shall be the number of palletized unit loads in the end item inspected lot. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5 in accordance with MIL-STD-105.

Examine	Defect
Finished dimensions	Length, width,or height exceeds specified maximum requirements.
Palletization	Pallet pattern not as specified. Interlocking of loads not as specified. Load not bonded with required straps as specified.
Weight	Exceeds maximum load limits.
Marking	Omitted, incorrect, illegible, of improper size, location, sequence, or method of application.

5. PACKAGING

5.1 Preservation-packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A. Twelve (12) hats of sizes 6 1/2, 6 3/4, 7 and 7 1/4 or eighteen (18) hats of sizes 7 1/2, 7 3/4, 8, 8 1/4 and 8 1/2 shall be nested together by placing the hat crown into each other. The nested hats shall be

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:52Z
Check the source to verify that this is the current version before use.

inserted in an end opening square style clear polyethylene bag of 0.0015 inch thickness (± 20 percent tolerance). The polyethylene bags shall be formed with heat sealed seams that are straight, continuous and parallel to each other and the formed edges of the bag. The end of the bag shall be heat sealed with the seal made as close as possible to the open end. A 1/4 inch diameter hole shall be made at one corner in each polyethylene bag to allow excess air to escape.

5.1.2 Commercial. Hats shall be packaged to afford adequate protection against physical damage during shipment from the contractor to the first receiving activity. The package and the quantity per package shall be the same as that normally used by the contractor for retail distribution.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

* 5.2.1 Level A. Thirty-six (36) hats, of one type and size only, packaged as specified in 5.1, shall be packed in a fiberboard shipping container, assembled, closed and reinforced conforming to type CF, class weather-resistant, variety DW, grade V15c, size 1A of MIL-B-17757. The fiberboard for the box liner shall conform to type CF, class domestic of MIL-B-17757. Level A packages of twelve (12) hats shall be packed upright, three in length, one in width, and one in depth within a shipping container. Level A packages of eighteen (18) hats shall be packed upright, two in length, one in width, and one in depth within a shipping container. The sides of each package of hats may be compressed so that they fit inside the container.

* 5.2.2 Level B. Thirty-six (36) hats, of one type and size only, packaged as specified in 5.1, shall be packed in a fiberboard shipping container, assembled and closed conforming to type CF, class domestic, variety DW, grade 200, size 1A of MIL-B-17757. The fiberboard for the box liner shall be type CF, class domestic of MIL-B-17757. Level A packages of twelve (12) hats shall be packed upright, three in length, one in width, and one in depth within a shipping container. Level A packages of eighteen (18) hats shall be packed upright, two in length, one in width, and one in depth within a shipping container. The sides of each package of hats may be compressed so that they fit inside the container.

* 5.2.3 Level C . Item, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall comply with the US Postal Manual, Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

* 5.2.4 Palletization. When specified (see 6.2) item, packed as specified shall be palletized on a 4-way entry pallet in accordance with load type Ia of MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with the bonding means C and D and F or G. Pallet patterns shall be in accordance with the appendix of MIL-STD-147.

The pallet shall be 4-way, type IV; type V, class 1, size 2; or type VIII, fabricated from wood groups I, II, III, or IV, grade A of NN-P-71, or 4-way, style 1, size A, type I, class 1 fabricated from wood groups specified, of MIL-P-15011. Interlocking of loads shall be effected by reversing the pattern

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:52Z
Check the source to verify that this is the current version before use.

of each course. If the container is of a size which does not conform to any of the patterns specified in MIL-STD-147, the pallet pattern used shall first be approved by the contracting officer.

* 5.3 Marking. In addition to any special marking required by the contract or order, unit packs, intermediate or exterior containers and palletized unit loads shall be marked in accordance with MIL-STD-129.

* 5.4 MIL-STD-2073. When specified, (see 6.2) Preservation and Packing shall be in accordance with MIL-STD-2073-1 and MIL-STD-2073-2.

* 6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

* 6.1 Intended use. The Type I hat is intended for wear by Navy enlisted men and the Type II hat is intended for wear by Naval Academy Midshipmen and NROTC Midshipmen.

* 6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.
b. Types and sizes required (see 1.2).
c. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. When first article inspection is required, (see 3.2) the item will be tested and should be a first article sample. The contracting officer should include specific instructions in acquisition documents regarding arrangement for examinations, quantity, and testing and approval.
e. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
f. When palletization is required (see 5.2.4).
g. When all packaging data is to be found in MIL-STD-2073-1 and MIL-STD-2073-2 (see 5.4).

6.3 Samples and patterns. For access to samples and patterns, address the procuring activity issuing the invitation for bids.

* 6.4 A suggested source for the nonwoven material specified in 3.3.3.1 is the Pellon Company, 119 West 40th St., New York, NY 10018, under their style number 3790.

* 6.5 Equal item. Prior to the use of an "or equal" item, the supplier shall submit the item with supporting data to the contracting officer for subsequent approval or disapproval by the responsible military agency.

6.6 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications,

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:52Z
Check the source to verify that this is the current version before use.

corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of the document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

* 6.7 Subject term (key word) listing.
Cap, enlisted
Cap, midshipman's
Cotton, 100%
Dixie Cup

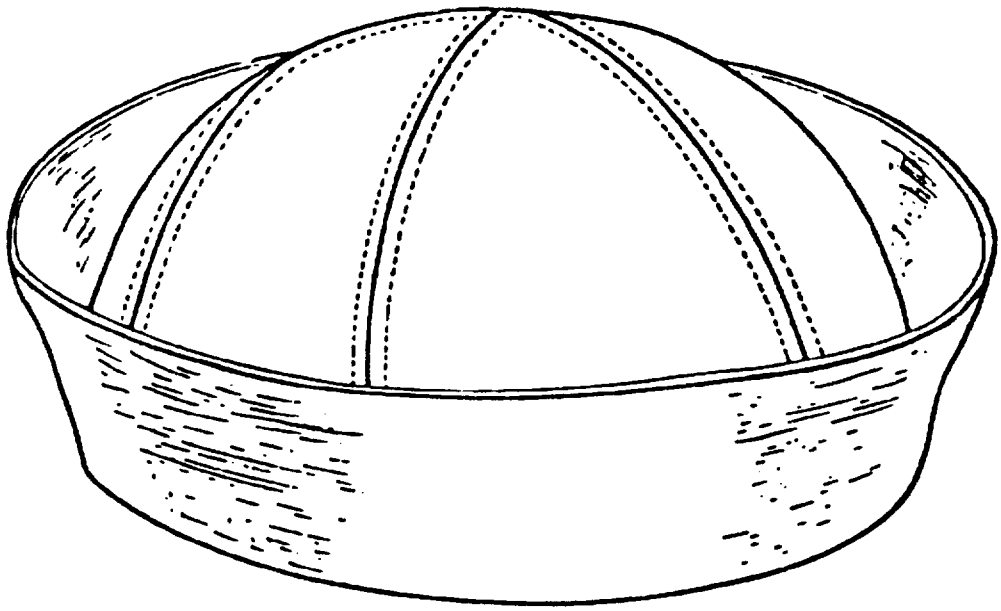
Custodian:
Navy - NU

Preparing Activity:
Navy - NU

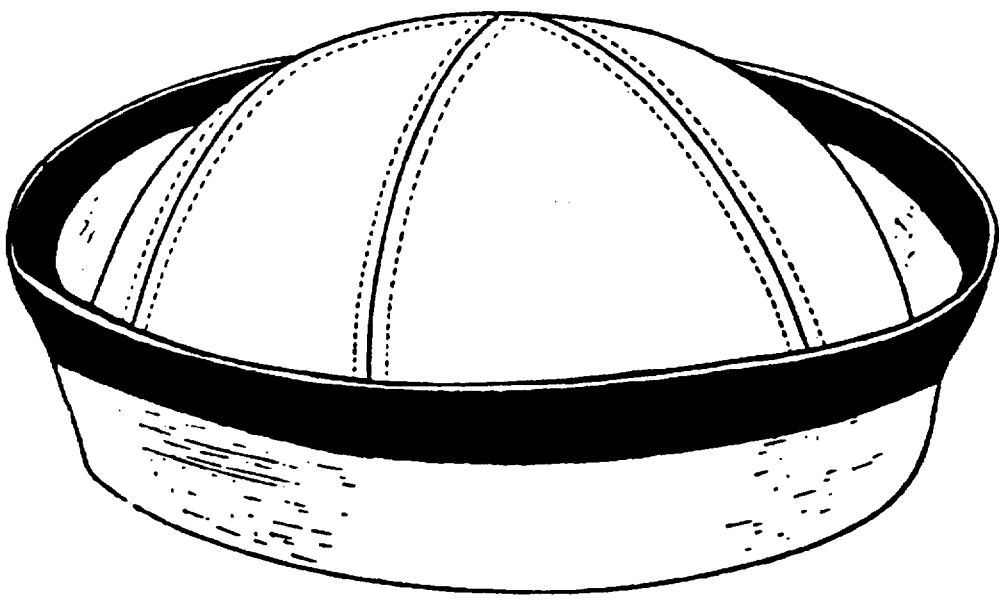
Review:
DLA-CT

Project No. 8405-N093

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:52Z
Check the source to verify that this is the currentversion before use.



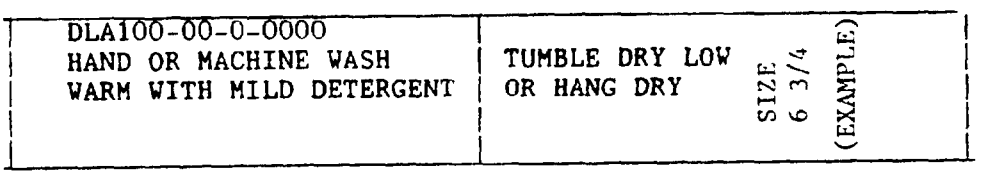
T-I ENLISTED MAN'S HAT



T-II MIDSHIPMAN'S HAT

Figure 1 - Hat, Service, White

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:52Z
Check the source to verify that this is the current version before use.



Fold
line

The finished length of the label shall be 2 to 2 1/8 inches.

Figure 2 - Hat, Service, White (label marking)

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:52Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:52Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE NAVY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12503 WASHINGTON D C.

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE NAVY

Navy Clothing and Textile Research Facility
21 Strathmore Road, (Code 50)
Natick, MA 01760-2490

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:52Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-H-87041B(NU)	HAT, SERVICE (WHITE)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil - Downloaded: 2026-10-26T04:52Z
Check the source to verify that this is the current version before use.